

Noah Zaranka

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SUMMARY

Full-stack web developer skilled in modern front-end frameworks, scalable backend systems, and cloud infrastructure. Focused on performance, security, and maintainability in building user-friendly applications.

Languages: TypeScript, JavaScript, Python, SQL, HTML/CSS, VBA

Frontend: React, Next.js, TailwindCSS, Three.js, GSAP, Vite, Responsive Design

Backend & Cloud: AWS (Lambda, DynamoDB, API Gateway, S3, CloudFront, Cognito, Amplify, Route53, CloudWatch), Serverless Architecture, REST APIs, Stripe

DevOps & Tools: Git, GitHub Actions, CI/CD Pipelines, Docker, FFmpeg, ImageMagick

EXPERIENCE

International Child Art Foundation

Washington, D.C.

Lead Developer

September 2023 – Present

- Architected and built a full-stack web application (Next.js, TypeScript, React, AWS) serving an international art competition
- Designed a serverless backend using Lambda, API Gateway, DynamoDB, S3, and CloudFront, optimizing for minimal data transfer and low operational cost
- Integrated Stripe payment processing with webhook-driven confirmation and automated database updates via API Gateway and DynamoDB
- Built a Lambda-based image processing pipeline generating thumbnails and display copies
- Implemented authentication with AWS Cognito using HttpOnly/Secure cookies and CSRF protection per OWASP best practices
- Created CI/CD workflows with GitHub Actions and AWS Amplify for zero-downtime releases with automated builds and cache invalidation
- Designed cost-control systems using CloudWatch alarms and Lambda triggers to automatically throttle services during traffic spikes

Developer Productivity Engineer

September 2023 – Present

- Recruited, onboarded, and mentored developer interns; managed GitHub access, AWS accounts, and task assignments
- Reviewed all pull requests, enforcing quality standards and providing guidance on architecture and best practices
- Led biweekly cross-functional meetings between development and design teams to align on requirements and feasibility

Web Systems & Reliability

August 2023 – Present

- Redesigned ICAF website frontend for full responsiveness; reduced load times by optimizing media with FFmpeg/ImageMagick and eliminating unnecessary preloading
- Managed site reliability across legacy HTML and modern Vite/React infrastructures, ensuring smooth migration and consistent uptime during the transition

PROJECTS

Puzzlr | Javascript, Three.js, Object-Oriented Programming

February 2023 – July 2023

- Developed an immersive 3D puzzle game using Three.js, allowing users to generate and solve puzzles from uploaded images, URLs, or preloaded selections in a dynamic 3D environment
- Created and adapted algorithms for puzzle generation, including a custom "Divided Squares" method and integration of Voronoi diagrams, programmatically generating SVG-based puzzle pieces with unique shapes
- Designed custom 3D assets and soundscapes, learning Blender for modeling and Ableton for dynamic audio feedback, enhancing the game's visual and auditory experience

Automated Video Pipeline | Python, Raspberry Pi, Bash, FFmpeg

May 2021 – September 2021

- Built a Raspberry Pi video processing pipeline that downloads, segments, and merges video streams using OCR-based scene detection, then auto-publishes highlights to YouTube

EDUCATION

Virginia Polytechnic Institute and State University

Blacksburg, VA

Bachelor of Science, Business Information Technology

Aug. 2015 – December 2018

- Concentration in Computer-Based Decision Support Systems: database management, web-based systems, systems analysis, and visual interface design